

The first Lazarev–Lieb phase sublevel retracts to the circle

A partial answer to Problem 6.3 of arXiv:1906.04417

Autonomous mathematical discovery run `fa_banach_001`

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Abstract

Frick and Superdock ask for the homotopy type of the space Y_n of smooth circle-valued paths with total variation at most πn , equipped with the L^1 topology. We determine the first nontrivial case. For every variation threshold $0 \leq L \leq \pi$, the normalized mean defines an antipodally equivariant strong deformation retraction of the corresponding phase space onto its circle of constant paths. In particular, $Y_1 \simeq_{\mathbb{Z}/2} S^1$ and $\text{coind } Y_1 = 1$. The proof includes the sharp endpoint $L = \pi$ and verifies continuity in the source’s coarse L^1 topology.

1 Source problem and scope

The source is F. Frick and M. Superdock, *A nonlinear Lazarev–Lieb theorem: L^2 -orthogonality via motion planning*, arXiv:1906.04417; J. Topol. Anal. 14 (2022), 569–585. Its Theorem 6.2 defines

$$Y_n = \{h \in C^\infty([0, 1]; S^1) : \|h\|_{W^{1,1}} \leq 1 + \pi n\}$$

with the topology induced by L^1 , proves $n \leq \text{coind } Y_n \leq 2n - 1$, and then asks in Problem 6.3 to determine the homotopy type of Y_n .

This $W^{1,1}$ -norm bound establishes an upper bound for the coindex of the space of smooth circle-valued functions with norm at most $1 + \pi n$:

Theorem 6.2. *For integer $n \geq 1$ let Y_n denote the space of C^∞ -functions $f: [0, 1] \rightarrow S^1$ with $\|f\|_{W^{1,1}} \leq 1 + \pi n$. Then*

$$n \leq \text{coind } Y_n \leq 2n - 1.$$

Proof. In the proof of Theorem 1.3 we constructed a $\mathbb{Z}/2$ -map $\beta_n: S^n \rightarrow Y_n$, which shows that $\text{coind } Y_n \geq n$. Let $f_1, \dots, f_n$ be chosen as in Theorem 6.1. Then the map $\psi: Y_n \rightarrow \mathbb{R}^{2n}$ given by $\psi(h) = (\int_0^1 f_j(x)h(x)dx)_j$ has no zero and is a $\mathbb{Z}/2$ -map. Thus ψ radially projects to a $\mathbb{Z}/2$ -map $Y_n \rightarrow S^{2n-1}$. A $\mathbb{Z}/2$ -map $S^{2n} \rightarrow Y_n$ would compose with ψ to a $\mathbb{Z}/2$ -map $S^{2n} \rightarrow S^{2n-1}$, contradicting the Borsuk–Ulam theorem. This implies $\text{coind } Y_n \leq 2n - 1$. $\square$

Problem 6.3. Determine the homotopy type of Y_n .

Because $|h| = 1$, the condition is exactly $\int_0^1 |h'(t)| dt \leq \pi n$. We solve the problem for $n = 1$, and more generally for every sublevel at or below this first critical threshold. We do not claim the all- n homotopy type.

2 Proof intuition

A circle-valued path of variation at most π fits in a closed semicircle. Its average cannot vanish: after rotating that semicircle to the upper half plane, cancellation would force the path to take only the two endpoints, which continuity forbids. More strongly, the mean direction lies strictly inside the path’s phase arc. No path value is therefore antipodal to the mean, so the principal argument relative to the mean is smooth. Multiplying that argument by $1 - s$ contracts the path to its mean direction without increasing variation. The mean is L^1 -continuous; a bounded-convergence argument handles the apparent branch-cut issue for the principal argument.

3 The deformation retraction

For $0 \leq L \leq \pi$, put

$$Y(L) = \left\{ h \in C^\infty([0, 1]; S^1) : \text{Var}(h) := \int_0^1 |h'(t)| dt \leq L \right\},$$

again with the L^1 topology. The involution is $h \mapsto -h$.

Lemma 1. *For every $h \in Y(\pi)$, the mean $m(h) = \int_0^1 h(t) dt$ is nonzero. If $u(h) = m(h)/|m(h)|$, then $-u(h) \notin h([0, 1])$.*

Proof. Lift h to a smooth real phase g , so $h = e^{ig}$. If $a = \min g$ and $b = \max g$, then

$$b - a \leq \text{Var}(g) = \text{Var}(h) \leq \pi.$$

If $a = b$, the assertions are immediate. Otherwise, continuity of g and the intermediate value theorem give

$$\Im(e^{-ia}m(h)) = \int_0^1 \sin(g(t) - a) dt > 0,$$

because $0 \leq g - a \leq \pi$ and the integrand is positive on a nonempty open set. Similarly,

$$\Im(e^{-ib}m(h)) = \int_0^1 \sin(g(t) - b) dt < 0.$$

Thus $m(h) \neq 0$, and it has an argument c strictly between a and b . Since $b - a \leq \pi$, neither $c + \pi$ nor $c - \pi$ belongs to $[a, b]$. Hence no value $e^{ig(t)}$ equals $-e^{ic} = -u(h)$. $\square$

Theorem 2. *For every $0 \leq L \leq \pi$, the inclusion of the constant paths $S^1 \hookrightarrow Y(L)$ is a $\mathbb{Z}/2$ -equivariant strong deformation retract. Consequently,*

$$Y_1 = Y(\pi) \simeq_{\mathbb{Z}/2} S^1, \quad \text{coind } Y_1 = 1.$$

Proof. For $h \in Y(L)$, Lemma 1 permits the smooth principal phase

$$\theta_h(t) = \text{Arg}(\overline{u(h)}h(t)) \in (-\pi, \pi).$$

Define, for $0 \leq s \leq 1$,

$$H_s(h)(t) = u(h) \exp(i(1-s)\theta_h(t)). \tag{1}$$

Then $H_0(h) = h$, $H_1(h) = u(h)$ is constant, and a constant path is fixed for every s . Moreover,

$$\text{Var}(H_s(h)) = (1-s) \int_0^1 |\theta'_h(t)| dt = (1-s) \text{Var}(h) \leq L,$$

so the homotopy stays in $Y(L)$.

It remains to check the topology carefully. The mean is a continuous linear functional on L^1 , and Lemma 1 makes normalization continuous on $Y(L)$. Suppose $h_k \rightarrow h$ in L^1 . Then $u(h_k) \rightarrow u(h)$ and $q_k = \overline{u(h_k)}h_k \rightarrow q = \overline{u(h)}h$ in L^1 . The compact set $q([0, 1])$ avoids -1 . On a fixed neighborhood of this set, the principal argument is uniformly continuous. Outside the set where q_k lies in that neighborhood, Markov's inequality makes the measure tend to zero; there the two arguments are merely bounded by π . Splitting the integral between these two sets proves $\theta_{h_k} \rightarrow \theta_h$ in L^1 . Formula (1) now proves joint continuity of H in (h, s) .

Finally, $u(-h) = -u(h)$, while $\overline{u(-h)}(-h) = \overline{u(h)}h$. Hence $H_s(-h) = -H_s(h)$: the retraction is equivariant. The antipodal circle has coindex one (and admits no equivariant map from S^2 by Borsuk–Ulam), so the same is true of Y_1 . $\square$

4 Why the same formula stops at the next threshold

The proof uses the bound $L \leq \pi$ exactly twice: to force a semicircle and to keep the normalized mean away from every antipodal path value. Both fail beyond the first threshold. More fundamentally, at variation 2π a full phase slip can be compressed into an interval of vanishing measure. Its exponential converges in L^1 , while any compatible real lift retains a 2π jump. Thus convexity of the space of real lifts does not descend to an L^1 -continuous contraction of Y_n for $n \geq 2$.

The accompanying attempt note records interval-moment/join, polynomial-phase, and phase-slip-cell approaches to the general problem. Each leaves a genuine continuous-selection or normal-form gap, so the result above is intentionally classified as partial.

5 Verification and novelty

The proof is analytic and exact. The included numerical script stress-tests the nonvanishing mean, antipode separation, and variation scaling on smooth random and boundary-near phase curves; it is not used as a substitute for the proof. The source page and source files are included locally.

A bounded search of the run indexes, the exact problem, the notation $\text{coind } Y_n$, and normalized-mean/homotopy phrases found no duplicate or published determination of Y_1 . Superdock’s 2021 thesis reproduces the general bounds and again states that exact coindex determination remains open. Novelty is plausible, not certified; a specialist should audit especially the L^1 -continuity paragraph in Theorem 2.

References

- [1] F. Frick and M. Superdock, *A nonlinear Lazarev–Lieb theorem: L^2 -orthogonality via motion planning*, J. Topol. Anal. 14 (2022), 569–585; arXiv:1906.04417.
- [2] M. Superdock, *Topics in topological combinatorics: Simplicial complexes, finite geometries, and the topology of circle-valued maps*, Ph.D. thesis, Carnegie Mellon University, 2021.